

Standard Tricks for Solving Standard Divisibilities

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1 Standard Tricks

- **Simplify the multiple:**

- $a \mid b \iff a \mid b + ac$ for any $c \in \mathbb{Z}$
- $a \mid b \implies a \mid bc$ for any $c \in \mathbb{Z}$
- $a \mid bc, \gcd(a, b) = 1 \implies a \mid c$

Given $x \mid y$, one can exploit these properties (most often by adding or subtracting multiples of x from y) to manipulate the given divisibility into nicer/simplified expressions.

- **Bounding:**

It is often extremely useful to apply the fact that $a \mid b \implies |a| \leq |b|$ or $b = 0$. It can also be useful to bound b/a .

And of course, always exploit properties of divisibility to your advantage whenever you can!

2 Problems

1. (AIME 1986 Q5) What is the largest positive integer n for which n^3+100 is divisible by $n+10$?
2. (ELMO SL 2024 N1) Find all pairs (n, d) of positive integers such that $d \mid n^2$ and $(n-d)^2 < 2d$.
3. (JBMO 2013 P1) Find all ordered pairs (a, b) of positive integers for which the numbers $\frac{a^3b-1}{a+1}$ and $\frac{b^3a+1}{b-1}$ are both positive integers.
4. (IMOSL 2021 N1) Find all positive integers $n \geq 1$ such that there exists a pair (a, b) of positive integers, such that $a^2 + b + 3$ is not divisible by the cube of any prime, and

$$n = \frac{ab + 3b + 8}{a^2 + b + 3}.$$

5. (Iran MO R2 2005 P1) Let $n, p > 1$ be positive integers and p be prime. We know that $n \mid p-1$ and $p \mid n^3-1$. Prove that $4p-3$ is a perfect square.
6. (APMO 2022 P1) Find all pairs (a, b) of positive integers such that a^3 is multiple of b^2 and $b-1$ is multiple of $a-1$.
7. (APMO 2013 P2) Determine all positive integers n for which

$$\frac{n^2 + 1}{[\sqrt{n}]^2 + 2}$$

is an integer.

8. (APMO 2002 P2) Find all positive integers a and b such that

$$\frac{a^2 + b}{b^2 - a} \quad \text{and} \quad \frac{b^2 + a}{a^2 - b}$$

are both integers.

9. (ELMO SL 2024 N4) Find all pairs (a, b) of positive integers such that $a^2 \mid b^3 + 1$ and $b^2 \mid a^3 + 1$.
10. (IMO 1998 P4) Determine all pairs (x, y) of positive integers such that $x^2y + x + y$ is divisible by $xy^2 + y + 7$.